Effects of Advanced Trauma Life Support® Training Compared to Standard Care on Adult Trauma Patient Outcomes (ADVANCE TRAUMA): A Cluster Randomised Trial

ALF Medicin 2026, projektmedel, Ny ansökan. Detta projekt har INTE fått ALF-medel under 2025

Martin Gerdin Wärnberg

Institutionen för folkhälsovetenskap, ME Intensivvård och Thoraxoperation

Basic information

Date of defense

2015-11-27

Current work place

Vårdenhet inom Region Stockholm

Work place within Region Stockholm

Karolinska universitetssjukhuset

Domicile of the project

Hospital or equivalent

Karolinska Universitetssjukhuset

Clinic/ward

Funktion PMI - ME Perioperativ medicin Solna

Postal address of the PI

Hospital

Karolinska universitetssjukhuset

Clinic

Funktion perioperativ medicin och intensivvård

Street address

Eugeniavägen 3

Postal code

17,165

City

Solna, Stockholm

Choose Priority allocation committee

Prioriteringsgrupp 6: Rörelseorganens sjukdomar och trauma, anestesi, intensivvård, kirurgi och odontologi

	Sum
2026	1099000
2027	1100000
2028	1100000
Sum	**3,299,000**

Clinical research time applied for

15

Other funding sources

Granted Vetenskapsrådet (2023-03128)
Effects of Advanced Trauma Life Support® Training Compared to Standard Care on Adult Trauma Patient Outcomes: A Cluster Randomised Trial

Granted sum
7550000 SEK
Date of decision
2023-10-31
Year
2029
Recipient
Martin Gerdin Wärnberg
Duration of the grant
2024-01-01
Principal investigator
Martin Gerdin Wärnberg
Granted Laerdal Foundation (2023-0297)
Effects of Advanced Trauma Life Support® Training Compared to Standard Care on Adult Trauma Patient Outcomes: Initiating A Cluster Randomised Trial

Granted sum
400000 SEK
Date of decision
2023-12-19
Year
2024
Recipient
Martin Gerdin Wärnberg
Duration of the grant
2024-01-01
Principal investigator
Martin Gerdin Wärnberg

Name of the relevant head of department/clinical director (or equivalent)

Name

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Approval of department head

I verify that the relevant clinical director (or equivalent) has approved the allocation of research time

Popular science description

Varje år dör 4.3 miljoner människor av trauma, alltså svår kroppskada som orsakas av olyckor eller våld. Ett bra primärt omhändertagande, alltså vård tidigt efter skadan, är avgörande för att rädda liv och minska sjuklighet. Flera kurser har tagits fram för att förbättra det primära omhändertagandet. Den mest kända kursen är Avanced Trauma Life Support ®, och mer än en miljon läkare över hela världen har utbildats i ATLS ® sedan 1978.

Trots att ATLS används i så stor omfattning finns det ingen forskning av hög kvalitet som visar att ATLS ® förbättrar utfallen hos patienter. Detta är problematiskt eftersom det betyder att många länder, inklusive Sverige, spenderar miljontals kronor varje år på att utbilda läkare i ATLS ® utan att veta om det leder till förbättrad överlevnad eller minskad sjuklighet hos patienter. Vi vill därför studera om ATLS ® förbättrar överlevnad och minskar sjuklighet hos vuxna traumapatienter.

Vi genomför projektet i Indien eftersom ATLS ® inte är standard där än. Vårt projekt blir det första att utvärdera effekten av ATLS ® på utfall hos patienter genom storskalig forskning av hög kvalitet. Vårt projekt är viktigt oavsett om resultaten är positiva eller negativa. Om vi visar att ATLS ® är effektivt bör liknande kurser införas i större skala, men om det inte är så behövs nya sätt att träna det primära omhändertagandet av traumapatienter.

Ethics review

Etik

Godkänd etik

Ethics applications

Effekter av träning i traumaomhändertagande på utfall hos vuxna traumapatienter: en klusterrandomiserad studie
Application no
2024-07547-01
Authority
Etikprövningsmyndigheten
Principal investigator
Martin Gerdin Wärnberg
Organisation
Karolinska Institutet
Decision
Godkänd
Decision date
2024-12-09

Co-applicants CV

Det är rubrikerna under respektive medsökande som är obligatoriska för ansökan

Co-applicants

Lovisa Strömmer

Capio AB

Current employment, duration of appointment, percentage of research, combined posts

Överläkare docent Capio St Görans Sjukhus (50%) och Karolinska Universitetessjukhuset (50%), ingen andel forskning i tjänsten

Connection to Region Stockholm

Arbetar på Karolinska Universitetssjukhuset

Other information relevant to the application

Anknytning Karolinska Institutet 20% Centrum för Hälsokriser

List of publications: the five most important and most relevant research papers, 2016-2024 (peer-reviewed)

1.Using the GAP score as a complement to the NISS score in identifying severely injured patients- A registry-based cohort study of adult trauma patients in Sweden. Granström A, Schandl A, Mårtensson J, Strömmer L. Injury. 2024 Sep;55(9):111709.
2.A systematic review of hemorrhage and vascular injuries in civilian public mass shootings. Nyberger K, Strömmer L, Wahlgren CM. SJTREM. 2023 Jun 19;31(1):30.
3.Patient and process factors associated with opportunities for improvement in trauma care: a registry-based study. Albajj H, Attegrim J, Strömmer L, Brattström O, Jacobsson M, Wihlke G, Västerbo L, Jonborg E, Gerdin Wernberg M. SJTREM 31:87, 2023.
4.Risk-adjusted mortality in severely injured adult trauma patients in Sweden. Strömmer L, Lundgren L, Ghorbani P, Troëng T. BJS Open 2022 Mar8;6(2) :zrac017. doi: 10.1093/bjsopen/zrac017.
5.Validation of the Norwegian survival prediction model in trauma (NORMIT) in Swedish trauma populations. Ghorbani P, Troëng T, Brattström O, Ringdal KG, Eken T, Ekbom A, Strömmer L. Br J Surg. 2020 Mar;107(4):381-390.

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Li Felländer-Tsai

Ortopedi

Current employment, duration of appointment, percentage of research, combined posts

Professor/överläkare, 33% forskning i tjänsten

Connection to Region Stockholm

Förenad anställning KI/Karolinska Universitetssjukhuset

Other information relevant to the application

List of publications: the five most important and most relevant research papers, 2016-2024 (peer-reviewed)

1: Escher C, Rystedt H, Creutzfeldt J, Meurling L, Hedman L, Felländer-Tsai L, Kjellin A. All professions can benefit - a mixed-methods study on simulation- based teamwork training for operating room teams. Adv Simul (Lond). 2023 Jul 17;8(1):18. doi: 10.1186/s41077-023-00257-0. PMID: 37460943; PMCID: PMC10351117.

2: Sumrein BO, Berg HE, Launonen AP, Landell P, Laitinen MK, Felländer-Tsai L, Mattila VM, Huttunen TT. Mortality following proximal humerus fracture-a nationwide register study of 147,692 fracture patients in Sweden. Osteoporos Int. 2023 Feb;34(2):349-356. doi: 10.1007/s00198-022-06612-7. Epub 2022 Nov 26.

PMID: 36435907; PMCID: PMC9852167.

3: Gerdin Wärnberg M, Berg J, Bhandarkar P, Chatterjee A, Chatterjee S, Chintamani C, Felländer-Tsai L, Gadgil A, Ghag G, Hasselberg M, Juillard C, Khajanchi M, Kizhakke Veetil D, Kumar V, Kundu D, Mishra A, Patil P, Roy N, Roy A, David S, Singh R, Solomon H, Soni KD, Strömmer L, Tandon M; Trauma life support training Effectiveness Research Network (TERN) collaborators. A pilot multicentre cluster randomised trial to compare the effect of trauma life support training programmes on patient and provider outcomes. BMJ Open. 2022 Apr 18;12(4):e057504. doi: 10.1136/bmjopen-2021-057504. PMID: 35437251; PMCID: PMC9016405.

4: Salonen A, Laitakari E, Berg HE, Felländer-Tsai L, Mattila VM, Huttunen TT. Incidence of femoral fractures in children and adolescents in Finland and Sweden between 1998 and 2016: A binational population-based study. Scand J Surg. 2022 Jan-Mar;111(1):14574969221083133. doi: 10.1177/14574969221083133. PMID: 35333132.

5: Möller H, Creutzfeldt J, Valeskog K, Rystedt H, Edelbring S, Fahlstedt M, Felländer-Tsai L, Abrandt Dahlgren M. Technology-Enhanced Learning of Human Trauma Biomechanics in an Interprofessional Student Context. Teach Learn Med. 2022 Apr-May;34(2):135-144. doi: 10.1080/10401334.2021.1893735. Epub 2021 Apr 1. PMID: 33792438.

Jag intygar att jag har uppdaterat mitt CV i Researchweb

Nobhojit Roy

Current employment, duration of appointment, percentage of research, combined posts

Director, WHO Collaborating Centre for Emergency, Critical and Operative Care, Program for Global Surgery & Trauma, The George Institute of Global Health, New Delhi, India. Duration: 4 years, 100% Research, Combined posts: Program Head of Global Surgery + Director WHO Collaborating Centre

Connection to Region Stockholm

Other information relevant to the application

List of publications: the five most important and most relevant research papers, 2016-2024 (peer-reviewed)

1. Roy N, Khajanchi M, Alty IG, Hamzah R, Aroke A, Banerjee N, Bhoi S, Chatterjee S, Soni KD, Gadgil A, Gururaj G, Jagnoor J, Joshi A, Joshipura M, Kamble J, Malhotra AK, Mehta S, Mock CN, Mohan R, Nathani P, Rawat R, Sarang B, Sharma MR, Sharma N, Sinha TP, Tewari P, Perez-Iglesias CT, Tripathi I, Leitz PTU, Raykar NP; TRAIN-Trauma India Symposium Collaborators. Consensus recommendations for acute trauma care & outcomes in LMICs from the transdisciplinary research, advocacy & implementation network for trauma in India. Indian J Med Res. 2024 Mar & Apr;159(3 & 4):274-284. doi: 10.25259/IJMR_2417_23. PMID: 39361792; PMCID: PMC11414786.

2. Pendleton AA, Sarang B, Mohan M, Raykar N, Wärnberg MG, Khajanchi M, Dharap S, Fitzgerald M, Sharma N, Soni KD, O'Reilly G. A cohort study of differences in trauma outcomes between females and males at four Indian Urban Trauma Centers. Injury. 2022 Sep 1;53(9):3052-8.

3. Amato S, Bonnell L, Mohan M, Roy N, Malhotra A. Comparing trauma mortality of injured patients in India and the USA: a risk-adjusted analysis. Trauma Surgery & Acute Care Open. 2021 Nov 1;6(1):e000719.

4. Roy N, Kizhakke Veetil D, Khajanchi MU, Kumar V, Solomon H, Kamble J, Basak D, Tomson G, von Schreeb J. Learning from 2523 trauma deaths in India-opportunities to prevent in-hospital deaths. BMC health services research. 2017 Dec;17:1-8.

5. Roy N, Gerdin M, Ghosh S, Gupta A, Kumar V, Khajanchi M, Schneider EB, Gruen R, Tomson G, Von Schreeb J. 30-day in-hospital trauma mortality in four urban university hospitals using an Indian trauma registry. World journal of surgery. 2016 Jun;40(6):1299-307.

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Research program

1. Scientific summary

Trauma is a massive global health issue. Many training programmes have been developed to help physician in the initial management of trauma patients. Advanced Trauma Life Support® (ATLS®) is the most popular of these programmes and have been used to train over one million physicians worldwide. Despite its widespread use, there are no controlled trials showing that ATLS ® improves patient outcomes. Multiple systematic reviews stress the need for such trials.

We will conduct a batched stepped wedge cluster randomised controlled trial with the aim to compare the effects of ATLS ® training with standard care on outcomes in adult trauma patients. Cluster will be hospitals in India. We will roll out the intervention of ATLS ® training to 30 clusters organised in six batches. Within each batch, clusters will be randomised to one of five implementation sequences. We expect a sample size of 4320 patients over five years.

This is a collaborative project between researchers and institutions in Sweden, India, and the United Kingdom. Our research will be important regardless of whether the results are positive or negative. If we show that ATLS ® improves patient outcomes then ATLS ® should be promoted to save lives and reduce morbidity. If we show that ATLS ® does not improve patient outcomes then this should impact how trauma life support training is delivered and new course concepts may be needed.

See attached Visual Abstract.

2. The question at issue/hypothesis

Each year, trauma - physical injury and the body's associated response - kills over four million people [1], is the top cause of burden of disease in young people aged 10-49 years [2], and costs trillions of dollars [3]. Advanced Trauma Life Support ® (ATLS ®) is the most established training programme aiming to help physicians in the initial management of trauma patients [4], but high-quality evidence from randomised trials showing that ATLS ® improves patient outcomes is missing [5–8]. We will conduct a cluster randomised trial to compare the effects of ATLS ® training with standard care on outcomes in adult trauma patients (ClinicalTrials.gov ID NCT06321419). The trial will be conducted in India and is funded by the Swedish Research Council (Dnr. 2023-03128) and the Laerdal Foundation (reg. no. 2023-0297), which covers including ten of the required 30 clusters. Here we apply for additional funding to facilitate recruiting the required number of clusters.

3. Research field overview

Most deaths from trauma occur within the first 24-48 hours [9]. Traumatic brain injury and exsanguination are the most common causes of trauma deaths [10,11]. Most preventable trauma deaths are caused by clinical judgement errors during initial resuscitation or early care including airway management and haemorrhage control, even though the deaths occur later during the hospital stay [10,12]. Several trauma life support training programmes have been developed to improve the early management of patients in the hospital by providing a structured framework for assessment and treatment [5–7,13,14]. The proprietary Advanced Trauma Life Support ® (ATLS ®) is the most established trauma life support training programme and more than one million physicians in over 80 countries have been trained in the programme since the first course in 1978 [4]. In the US and many other countries training in ATLS ® is virtually mandatory for trauma care physicians [15].

There are three randomised studies showing that ATLS ® improves knowledge and clinical skills [16–18], but there are no randomised controlled trials or high-quality quasi-experimental trials indicating that ATLS ® improves patient outcomes [5–8,14]. Since 2020, we have conducted a series of systematic searches in collaboration with the Karolinska Institutet University Library to identify studies on the impact of ATLS ® training on patient outcomes, and our latest search was conducted on January 22, 2025. We have screened

a total of 10,429 records and estimated a pooled effect of ATLS ® training on mortality from 12 heterogeneous (I^2 0.9) observational studies as a risk ratio of 0.76 (95% CI 0.57; 1.01) (Figure 1) [19–30]. These studies indicate that ATLS ® may reduce mortality, but that the quality of the evidence is low.

4. Objectives and effects

We propose an ambitious but high-yield project that will result in the first robust evidence on the effects of ATLS ® on patient outcomes since the programme was introduced. We advance the trauma research frontier by conducting large-scale research on a complex system intervention and move beyond the prevailing focus on observational research [19–30]. Systematic reviews call for cluster randomised or quasi-experimental trials in settings where ATLS ® or similar programmes are not routinely taught [5–7]. Our findings will be important regardless of whether they are positive or negative. If ATLS ® improves patient outcomes it should be further promoted. If ATLS ® does not improve patient outcomes then trauma life support training needs to change.

5. Project description

This project was developed jointly by the parties participating in the Trauma life support training Effectiveness Research Network (www.tern.network), funded by the Swedish Research Council (Dnr. 2020-03779). We report our methods according to the Consolidated Standards Of Reporting Trials (CONSORT) extension for stepped-wedge cluster randomised controlled trials [31]. The trial is registered with ClinicalTrials.gov, number NCT06321419, and approved by the relevant regulatory and ethical review authorities.

Design

We will conduct a batched stepped-wedge cluster randomised controlled trial in India (Figure 2). We conduct this trial in India because of 1) physicians in India are not routinely trained in ATLS ® or similar programmes and 2) our established collaboration with Indian institutions and experience in conducting multicentre studies in this setting (www.titco.org). The stepped-wedge trial is a uni-directional cross-over trial but the time point when clusters cross-over from standard care to the intervention is randomised [32]. We will roll out the interventions to 30 clusters over six batches, so there will be five clusters in each batch. Each cluster will be at least one unit of physicians performing initial resuscitation of trauma patients in the emergency department of tertiary hospitals in India. The number of units that we will train in each hospital will depend on the sizes of these units and the volumes of patients that they see.

If more than one unit is trained in the same hospital these units will be considered one cluster for the purpose of randomisation. We choose this approach for two reasons: 1) it will not be logistically or financially feasible to train all physician in a given hospital; and 2) we need to balance cluster size with the number of clusters. The clusters in each batch will be randomised to one of five implementation sequences, with one hospital randomised to each implementation sequence. All clusters will transition through three phases, first a standard care phase, then a one month transition phase during which the training is delivered, and finally an intervention phase, for a total of 13 months. The implementation sequence determines how long the phases of standard care and intervention are. Patient participants will be followed up for a total of three months.

Design justification

We use the cluster randomised design because the intervention cannot be randomised at the individual patient level. We use the stepped-wedge design for two reasons. First, this design is statistically more efficient than the parallel cluster design when the number of clusters is limited [33]. In this trial, the number of clusters is limited because of the costs associated with ATLS ® training and the available slots for ATLS ® training in India. Second, the stepped-wedge design is likely to enhance participation and engagement because all clusters receive the intervention. The batched stepped-wedge design further improves feasibility as it does not require all clusters to start at the same time, and it is robust to potential delays in cluster recruitment [34].

Participants

Our trial has eligibility criteria on three levels: hospitals, clusters and patient participants. Hospital will be the unit of randomisation. To be eligible, **hospitals** need to admit at least 400 trauma patients per year and provide surgical and orthopaedic emergency services around the clock. **Clusters** will be one or more units of physicians providing initial trauma care in the emergency department of included hospitals. These units must admit or refer/transfer for admission at least 12 patients with trauma per month. **Patient participants** will be adult trauma patients who are admitted at participating hospitals. We define a history of trauma using the International Classification of Diseases chapter 20.

Intervention and control treatment

The intervention will be ATLS ®[4] training. This is a proprietary 2.5 day course teaching a standardised approach to intial treatment and resuscitation, triage and interfacility transfers of trauma patients using the concepts of a primary and secondary survey. The programme was developed by the Committee of Trauma of the American College of Surgeons. **The control will be standard care**, meaning no formal trauma life support training. During the transition phase, we will train the physicians that initially resuscitate and provide trauma care during the first hour after patient arrival at the emergency department. These physicians can be casualty medical officers, surgical residents, or emergency medicine residents, depending on the setup at each participating centre.

Outcomes

Primary outcome

The primary outcome will be in-hospital mortality within 30 days of arrival at the emergency department. Clinical research coordinators will extract information on death from patient hospital records. If the patient has been transferred to another hospital, the clinical research coordinators will collect data on this outcome by calling the patient or a patient representative, or by contacting the hospital to which the patient was transferred. Data on this outcome will be collected continuously during the trial.

Secondary outcomes

- All cause mortality within 24 hours, 30 days and three months of arrival at the emergency department.
- Length of emergency department stay.
- Length of hospital stay.
- Intensive care unit admission.
- Length of intensive care unit stay.
- Adherence to ATLS ® principles during initial patient resuscitation, up to one hour after the physician has first seen the patient. This assessment will be done using a 14 item checklist covering the key steps of the ATLS ® primary survey, which was modelled based on previous work on ATLS ® adherence [35]. We will consider completion of all 14 steps as 100% adherence. The clinical research coordinators collecting the data will be trained by the trial team to do this, prior to the start of the trial. We will collect this data by observing the care of a random sample of patients.
- Quality of life within seven days of discharge, and at 30 days and three months of arrival at the emergency department, measured by the official and validated translations of the EQ5D3L. Data on this outcome will be collected in person if the patient is still in hospital, or by phone if the patient has been discharged.
- Disability within seven days of discharge, and at 30 days and three months of arrival at the emergency department, assessed using the WHO Disability Assessment Schedule 2.0 (WHODAS 2.0). Data on this outcome will be collected in person if the patient is still in hospital, or by phone if the patient has been discharged.
- Return to work at 30 days and three months after arrival at the emergency department. Data on this outcome will be collected in person if the patient is still in hospital, or by phone if the patient has been discharged.

Data analysis and statistics

General principles

We will conduct all analysis by modified intention to treat. Clusters and observations within clusters will be considered exposed to the intervention after the date at which the cluster was scheduled to transition. All data will be included with the exception of the transition phases. We will not adjust for multiplicity of analyses because none of the secondary outcomes will be singularly more important. However, all secondary outcomes will be interpreted with due consideration for how all are affected by the intervention without putting any undue emphasis on a single outcome that might be statistically significant but where all others appear to have remained unchanged. We will use a two-sided significance level of 5% and estimate 95% confidence intervals. Subgroup analyses will include age across the groups older adolescents (15-19 years), young adults (20-24 years), adults (25-59 years), and older adults (60 years and older); sex; and the clinical cohorts blunt multisytem trauma, penetrating trauma, and severe isolated traumatic brain injury.

Analysis models

There are a number of requirements for the analysis model. Firstly, all analysis will consider the clustered nature of the design. Secondly, as the trial has only 30 clusters, it will be essential that the model allows for a correction due to the small number of clusters. Thirdly, as the design is a stepped-wedge study, we will adjust for temporal confounding using categorical effects for period of the study (month). For binary outcomes, a mixed effects binomial regression with a logit link will be used to estimate the odds ratio; and a binomial model with identity link used to estimate the risk difference. These models will be fitted using residual pseudo-likelihood estimation based on linearization with subject-specific expansion (RSPL). If the binomial model with the identity link does not converge then only a odds ratio will be reported. We will include fixed effects for period and a fixed effect for intervention exposure. To correct the potential inflation of the type I error rate due to small number of clusters, a correction for a small number of clusters will be applied, but the correction that will be selected will be based on the best available evidence available closer to the time. In a sensitivity analysis we will explore if models with more complicated correlation structures are a better fit to the data. These models are not being used as our primary analysis models as there is limited understanding as to when such models will converge and how to choose between the various different correlation structures which might be plausible. To allow for the randomisation by batches, a different secular trend will be included for each batch (interaction between batch and period). For continuous, count and prevalence outcomes similar model-based approaches will be used but with appropriate links and distribution functions, using transformations where appropriate.

Power analysis

To detect a reduction in all-cause mortality within 30 days from 20% under standard care to 15% after ATLS ® training as significant at the 5% level, with ~90% power across different combinations of cluster autocorrelations and intra-cluster correlations [36], we need a total sample size of 4320 patients across 30 clusters (Figure 3). This mortality effect is a conservative estimate equalling a risk ratio of 0.75, which would be both clinically important and consistent with our pilot study and updated systematic review. We assume that each cluster will contribute approximately 12 observations per month to the analysis, based on our previous work.

Time table

The necessary regulatory and ethical approvals have been obtained and we expect to start data including patients during February 2025. The trial will then run for four years, 2025 included. During the following years we will enrol clusters and obtain approvals for each subsequent batch. The main risks and our corresponding mitigation plans are:

- **delays in obtaining approvals**: be proactive and start necessary processes, for example submissions to ethical review boards, long before the study is planned to start in each centre;
- **delays in cluster recruitment**: approach potential clusters long before the intended study start, leveraging our considerable existing networks to (the hospitals that will participate in the first batch are already finalised);
- **lower than expected enrolment rates**: perform careful on-site evaluations before a cluster is formally enrolled to avoid lower than expected enrolment rates.

There is also the risk of unforeseen global events like the COVID-19 pandemic. While we cannot mitigate this risk we have limited its potential impact on our trial by adopting the batched stepped-wedge design, which makes it feasible to pause and resume the trial between batches without compromising its quality.

6. Preliminary results

We conducted a pilot cluster randomised controlled trial (ClinicalTrials.gov NCT05417243) between April 2022 and February 2023 to assess the feasibility of a full scale trial. We published the protocol for this pilot study [37] and the results are available as a preprint [38]. Our pilot study enrolled 375 patients from seven hospitals across India and shows that it is feasible to conduct the proposed trial with a high inclusion rate and low missingness in key variables. To involve patients and the public in the planning of this trial we conducted semi-structured interviews with trauma patients, caregivers, and community representatives. The aim of these interviews was to understand their views on the trial and important outcomes and they showed high acceptability of our research and emphasised the importance of better recovery before discharge and reduced disability. The interviews also highlighted return to work as an important outcome. The results are also available as a preprint [39].

7. Ethical issues

Our trial has received ethical approval from the Swedish Ethical Review Authority, the George Institute for Global Health in India, and all hospitals participating in the first batch.

There are three main ethical issues with our research:

- Trauma patients constitute a vulnerable population because they often arrive in a state of emotional and physical distress with injuries and pain from injuries affecting their level of consciousness. They also often lack the protection of a proxy decision maker.
- It will not be possible for patients to opt out from being subjected to the intervention, because the intervention is delivered at the cluster level and involves training physicians in trauma life support. Therefore, as physicians cannot be expected to temporarily forget their training.
- We will apply for a waiver of informed consent from the ethical review boards at the participating hospitals. We do this because the target population will often have reduced decisional capacity at the time of data collection and the nature of the intervention is such that they can not refuse receiving it.

Regulations on medical research in Sweden and India allow such a waiver when the research is expected to generate knowledge that cannot be generated through research using informed consent, or the research is expected to directly benefit research participants. If the research is not expected to directly benefit research participants it is still allowed if the research is expected to generate knowledge that can directly benefit the patient population to which the participant belongs, and the research involves minimal risk.

Our research measure up to these issues for the following reasons. First, it cannot be conducted using informed consent because the cohort of patients with severe trauma is the group most likely to benefit from improvements in early management and it is therefore crucial to include this population. Second, the research can directly benefit the participants during the intervention phase if ATLS is found to improve patient outcomes. The procedures that we will use to collect data will be direct observation of care, routine physical examinations, questionnaires, and extraction of already collected data from patient records, which are often seen as involving only minimal risk.

8. Patient benefit/significance for the health service

A. Planned implementation:

In Sweden, and thus in Region Stockholm, ATLS ® is currently recommended as part of residency training in surgery, neurosurgery, orthopaedics, and anaesthesiology, despite the lack of high-quality evidence showing that it improves patient outcomes. In addition to the lack of evidence on its effectiveness, ATLS ® has been criticised for high costs, infrequent updates, lack of contextual adaption, limited evidence on the effectiveness of its individual components, and how some of its concepts, like the categorisation of haemorrhagic shock, lack clinical and theoretical support. Because of these concerns, there have been repeated calls for large-scale clinical trials, but so far none have been conducted.

This project is the first to challenge this longstanding trauma care dogma on this level, with direct clinical relevance for the training of trauma care physicians. A positive result reinforces ATLS ® use and should guide its optimisation internationally as well as in Sweden and Stockholm. It would directly support the Swedish recommendations to train residents in ATLS ®, and justify national and regional spending on this training. A positive result would promote further research on the effectiveness of individual components of ATLS ®, whom it works for and why, and how it best is implemented in different settings. It would also allow for studies on its cost-effectiveness. Within five and ten years, ATLS ® would remain the standard of care for trauma care physicians.

A negative result calls for new approaches to trauma life support training. The fact that this trial is conducted in India should not stop a negative result from spurring a healthy national debate on how we train trauma care physicians, especially considering that the current edition of ATLS ® states how "ATLS training in a developing country has resulted in a decrease in injury mortality" as a justification for its impact. Within five years, we expect to see confirmatory trials in other settings, including comparative trials with other training programmes in settings where ATLS ® is currently the standard. Within ten years, we expect other training approaches, better adapted to the realities of trauma care in different settings, to be explored.

B. Clinical advocacy:

The main applicant **Martin Gerdin Wärnberg** is clinically active corresponding to 70% of full time as a resident in anaesthesia and intensive care. His research time is covered by other grants. **Lovisa Strömmer** is clinically active corresponding to 80% of full time as a senior consultant in surgery. She is actively engaged in quality improvement initiatives for trauma care and is a member of local and national clinical committees for trauma care. This grant will cover 5% of her research time. **Li Felländer-Tsai** is clinically active corresponding to 33% of full time as a professor of orthopaedic surgery and senior consultant at the Karolinska University Hospital. She is involved with multiple national and international orthopaedic societies, and she is the director of the Center for Advanced Medical Simulation and Training (CAMST) at the Karolinska University Hospital. This grant will cover 5% of her research time. **Nobhojit Roy** is a trauma surgeon and the Director of the WHO Collaborating Centre for Emergency, Critical and Operative Care, Program for Global Surgery & Trauma at the George Institute of Global Health, New Delhi, India. He is also affiliated with Karolinska Institutet. This grant will cover 5% of his research time. Collectively, the applicants ensures that the findings of this project can be disseminated in the relevant fora and implemented in clinical practice locally, nationally, and internationally. In addition to the applicants, this grant will cover the recruitment of a clinically active doctoral student, corresponding to 40% of full time.

C. Generalisability of anticipated findings:

Several systematic reviews have called for large-scale trials of ATLS ® in settings where it is not currently the standard of care, because it would not be possible to conduct this research in a setting where most, if not all, physicians are already trained in ATLS ®. The generalisability of our findings need to be considered in this context, but we argue that our findings will be applicable to settings outside of the studied setting. The findings are likely to be directly applicable to other settings with similar health care systems and current level of training in trauma care, primarily in low- and middle-income countries. However, we argue that the direction of the effects will be generalisible to settings with more mature trauma care systems, including Sweden, and can either support or refute current observational research.

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AI i ansökan

This application was written in the Cursor Integrated Development Environment (IDE), which includes integrated AI assistance. This assistance was used to help structure parts of the application and check spelling and grammar.

Bilagor

visual-abstract.png
Visual abstract
figure-1.png
Figure 1: Summary of the systematic review on the effect of Advanced Trauma Life Support® (ATLS®) on mortality, updated for this application. Abbreviations: CI Confidence Interval, RR Risk Ratio.
figure-2.png
Figure 2: Trial design. The lightly colored blue area in the middle of the figure indicates the period of the trial covered by the ALF project grant now applied for. Lines represent the duration of patient enrolment across clusters and phases. Clusters will be sequentially allocated to a batch based on when they enter the study. Within each batch clusters will then be randomised to an intervention implementation sequence.
figure-3.png
Figure 3: Power curves for different combinations of cluster autocorrelations (CAC) and intra-cluster correlations (ICC), to detect a reduction in the primary outcome of in-hospital mortality within 30 days from 20% under standard care to 15% after ATLS® training as significant at the 5% level.

Research group and competence

Presentation of the research group and description of competence

Martin Gerdin Wärnberg is an Associate Professor of Clinical Epidemiology at KI and resident in anaesthesia and intensive care at Function Perioperative Medicine and Intensive Care at the Karolinska University Hospital in Solna. He leads the Improving Trauma Systems team at the Department of Global Public Health, with two PhD students and one affiliated postdoctoral researcher. Martin has led multicentre observational and interventional trauma research in India for over 10 years, and he has experience of working with Swedish trauma registry data. He will coordinate the project and oversee data management, analysis and reporting. Martin has previous experience from conducting the pilot cluster randomised controlled trial leading up to this full-scale trial. His planned research time in this project is 30%. **Li Felländer-Tsai** is a Professor of Orthopaedic Surgery at KI and Senior Consultant at the Karolinska University Hospital. Li will contribute her experience of delivering and studying training interventions. Her planned research time in this project is 5%. **Lovisa Strömmer** is a trauma surgeon and Senior Consultant in Surgery at the Karolinska University Hospital and Capio St Göran Hospital, as well as Associate Professor of Surgery at KI. Lovisa will contribute with her substantial clinical trauma expertise and experience in researching trauma outcomes. Her planned research time in this project is 5%. **Nobhojit Roy** is a trauma surgeon and the Director of the WHO Collaborating Centre for Emergency, Critical and Operative Care, Program for Global Surgery & Trauma at the George Institute of Global Health, New Delhi, India. He is also affiliated with Karolinska Institutet. His planned research time in this project is 5%. In addition to the applicants, this project will also recruit a clinically active doctoral student, with 40% planned research time.

Your role in the project

I am the principal investigator of this project and am responsible for the overall design and conduct of the trial. I was the lead applicant for the grants from the Swedish Research Council and Laerdal Foundation that have been awarded to fund this project so far.

As the principal investigator, I am part of all trial committees, which have all been established. My roles in these committees are:

- I am a member of the Trial Team that runs the day to day operations of the trial;
- I chair the Trial Management Group that is responsible for the overall management of the trial and includes 17 members from institutions in Sweden, India and the UK; and
- I participate in the meetings of the Joint Trial Steering and Data Monitoring Committee, which is the trial's oversight body and is chaired by an external member and includes two external clinical experts, an external biostatistician, and a lay representative.

My time in this project is 30% of full time, and is funded by the Swedish Research Council. The other projects being run within the group of co-applicants focus on trauma quality improvement initiatives, with completed data collection or based on trauma registry data.

Age of PI

Under 65 years

Resources, network, infrastructure and budget

1. Resources for implementation of the project - in terms of personnel, materials and finance

The resources required to implement the project involves the research time of the co-applicants and the doctoral student, trial management and data collection, and delivering of the intervention. Trial management and data collection is the responsibility of the George Institute for Global Health in India and involves a full time team of one project manager, one clinical research associate, data managers, and one clinical research coordinator per cluster.

2. Specifiy and motivate the items in the budget

	Beskrivning av kostnaderna	År 2026	År 2027	År 2028
Personnel costs, Region Stockholm employees|SWE:Personalkostnad Region Stockholm	Research time corresponding to 5% of full time for co-applicants Lovisa Strömmer and Li Felländer-Tsai, and research time corresponding to 40% of full time for clinically active doctoral student. The costs includes a yearly increase of 2%.	544000	566000	577000
Prsonnel costs, other employees|SWE:Personalkostnad övriga	Research time corresponding to 5% of full time for co-applicant Nobhojit Roy. The costs includes a yearly increase of 2%.	94000	98000	100000
Other costs|SWE:Övriga kostnader	Project management, intervention implementation and data collection in by the George Institute for Global Health in India.	461000	436900	423000
Total costs for this application|SWE:Totala kostnader för denna ansökan		1099000	1100000	1100000
Total costs for the project|SWE:Projektets totala kostnad	Includes 30% salary for the principal investigator, costs for the trial statistician, additional international collaborators, office space and IT-fees. The difference between the funding available in this call and the total costs are partly covered by other grants and partly applied for from other sources.	3 470 000	3 500 000	3 520 000

2.1 Intended use of ALF Medicine project funding

The funding from this grant will be used to cover the research time of the co-applicants and the doctoral student, and to recruit clusters in the remaining batches. We anticipate that this will be completed within the three year time frame of this grant. Funding for batch 1 and 2 has already been secured through grants from the Swedish Research Council and the Laerdal Foundation.

3. Cooperation partners and networks

This trial is an already established collaborative effort between a total of 22 members from twelve institutions: All India Institute of Medical Sciences, India; Grant Govt. Medical College, India; Institute of Post Graduate Medical Education & Research, India; Jawaharlal Institute of Postgraduate Medical Education and Research, Puducherry, India; Karolinska Institutet; Karolinska University Hospital; King Edward Memorial Hospital, India; Maulana Azad Medical College, India; Queen Mary University of London, UK; Stellenbosch University, South Africa; The George Institute for Global Health, India; and University of Birmingham, UK. Key collaborators include **Vivekanand Jha**, Professor and the Executive Director of the George Institute for Global Health India, contributes his expertise in managing similar large scale trials in India. **Karla Hemming**, Professor of Biostatistics at the University of Birmingham, is an international expert in stepped-wedge trials and provides input on design and analysis. She authored the CONSORT extension for stepped-wedge trials.

5. Research environment/infrastructure

This trial is conducted in the collaborative research environment generated by the participating institutions. The trial groups and committees are established and meet regularly to oversee and discuss the trial. The institutions to which the main applicant is affiliated provide conducive environments for high quality research. The Department of Global Public Health at KI is a highly multidisciplinary department with a strong focus on research and education in public and global health. There are twelve research groups and about 110 doctoral students at the department, with regular research seminars, workshops and courses, providing ample opportunities for exchange of knowledge and collaboration, and for the prospective doctoral student to interact with other students and researchers. Martin's team at the Department of Global Public Health involves both national and international researchers, who meet at least monthly to discuss

ongoing research. Function Perioperative Medicine and Intensive Care (PMI) at the Karolinska University Hospital has strong clinical research activities and infrastructure, including biostatistics support, available for the project. This project expands the local and national research projects conducted at PMI with a large-scale international trial. The George Institute for Global Health is an international research organisation with extensive experience in conducting large scalse clinical trials, with well developed infrastructure for trial management, including monitoring and adherence to clinical trial regulations, and data collection.

Gender perspective

Description of the gender perspective

The sex and gender perspectives are integral parts of our research. We have shown that almost 80% of both patients who present and patients who are admitted with trauma in the setting of our proposed research are male. We have also shown that female and male patients have similar odds of early and late mortality after adjusting for potential confounders. In yet unpublished data, we find that old-aged females (60 years and above) and middle-aged men (33-59 years) report the lowest health status, while middle-aged females had the highest odds of reporting problems across the functional domains. In qualitative interviews with trauma patients after discharge from hospital we find that females and males face different challenges, especially regarding self care and daily activities.

These previous findings indicate that we can expect outcomes, and particularly non-mortality outcomes, to differ between females and males. In our proposed trial we therefore take account of sex and gender perspectives in two ways. First, we will adjust for sex in our fully adjusted covariate analysis. This will allow us to identify and discuss the effect of patient sex on the trial outcomes. Second, we will perform prespecified subgroup analyses in females and males respectively. These subgroup analyses will allow us to explore potential heterogeneity in the intervention effect in females and males. These subgroup analyses will unfortunately need to be explorative and hypothesis generating, as it is beyond the limits of the funding available within this call for these analyses to achieve the same power as the main analysis.

CV för huvudsökande

Current employment

Universitet - Karolinska Institutet - Institutionen för folkhälsovetenskap (100 %)

Andel forskning i tjänsten

Kombinationstjänst?

Nej

Regioner - Region Stockholm - Karolinska Universitetssjukhuset - Funktion PM - ME Perioperativ medicin Solna (100 %)

Andel forskning i tjänsten

Kombinationstjänst?

Nej

Connection to Region Stockholm

The main and two of the co-applicants are employed by and clinically active at the Karolinska University Hospital.

Other matters that are relevant to the application

Counted from the date of defence, the main applicant can deduct a total of at least 60 months of active research time for parental leave and clinical work.

Applicant's scientific qualifications

The 10 most important and most relevant research papers for the project, 2017-2025

first authorships

1

articles as first author

last authorships

7

articles as last author

1. Patient and process factors associated with opportunities for improvement in trauma care: a registry-based study.
   Albaaj H, Attergrim J, Strömmer L, Brattström O, Jacobsson M, Wihlke G, Västerbo L, Joneborg E, Gerdin Wärnberg M.
   Scand J Trauma Resusc Emerg Med 2023:31(1):87.
   Web of Science® citations 5
   PMID 38012791 10.1186/s13049-023-01157-y
2. Facilitators and barriers impacting in-hospital Trauma Quality Improvement Program (TQIP) implementation across country income levels: a scoping review.
   Kapanadze G, Berg J, Sun Y, Gerdin Wärnberg M.
   BMJ open 2023:13(2):e068219.
   Web of Science® citations 8
   PMID 36806064 10.1136/bmjopen-2022-068219
3. Perceived usefulness of trauma audit filters in urban India: a mixed-methods multicentre Delphi study comparing filters from the WHO and low and middle-income countries.
   Berg J, Alvesson HM, Roy N, Ekelund U, Bains L, Chatterjee S, Bhattacharjee PK, David S, Gupta S, Kamble J et al [Petzold M Nr. 17] [Gerdin Wärnberg M Nr. 23].
   BMJ Open 2022:12(6):e059948.
   Web of Science® citations 2
   PMID 35680271 10.1136/bmjopen-2021-059948
4. A pilot multicentre cluster randomised trial to compare the effect of trauma life support training programmes on patient and provider outcomes.
   Gerdin Wärnberg M, Berg J, Bhandarkar P, Chatterjee A, Chatterjee S, Chintamani C, Felländer-Tsai L, Gadgil A, Ghag G, Hasselberg M et al;
   Trauma life support training Effectiveness Research Network (TERN) collaborators.
   BMJ Open 2022:12(4):e057504.
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   PMID 35437251 10.1136/bmjopen-2021-057504
5. Measuring socioeconomic outcomes in trauma patients up to one year post-discharge: A systematic review and meta-analysis.
   David SD, Aroke A, Roy N, Solomon H, Lundborg CS, Gerdin Wärnberg M.
   Injury 2022:53(2):272-285.
   Web of Science® citations 14
   PMID 34706829 10.1016/j.injury.2021.10.012
6. Towards definitions of critical illness and critical care using concept analysis.
   Kazidule Kayambankadzanja RKK, Schell COS, Gerdin Wärnberg MGW, Mollazadegan HM, Holmberg MH, Molsted Alvesson HMA, Baker TB.
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   Fulltext
7. 'Coming home does not mean that the injury has gone'—exploring the lived experience of socioeconomic and quality of life outcomes in post-discharge trauma patients in urban India.
   David S, Roy N, Stålsby Lundborg C, Gerdin Wärnberg M, Solomon H.
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Link
8. The transfer of clinical prediction models for early trauma care had uncertain effects on mistriage.
Henriksson M, Saulnier DD, Berg J, Gerdin Wärnberg M.
J Clin Epidemiol 2020:128:66-73.
Web of Science® citations 2
PMID 32835888 10.1016/j.jclinepi.2020.08.014
9. Prevalence of a definitive airway in patients with severe traumatic brain injury received at four urban public university hospitals in India: a cohort study.
Khajanchi MU, Kumar V, Wärnberg Gerdin L, Soni KD, Saha ML, Roy N, Gerdin Wärnberg M.
Inj Prev 2019:25(5):428-432.
Web of Science® citations 5
PMID 29866716 10.1136/injuryprev-2018-042826
10. Mortality following surgery for trauma in an Indian trauma cohort.
Mansourati M, Kumar V, Khajanchi M, Saha M, Dharap S, Seger R, Warnberg M.
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Original research 2020-2025 - peer-reviewed

1. Patient and process factors associated with opportunities for improvement in trauma care: a registry-based study.
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2. Profile and triage validity of trauma patients triaged green: a prospective cohort study from a secondary care hospital in India.
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